# Sourav Ahmed

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## SUMMARY

**Full-Stack Software Engineer • AI/ML Enthusiast • Open Source Contributor** with 3+ years building scalable applications and distributed systems at high-growth startups. Expert in JavaScript/TypeScript, real-time systems, and cloud infrastructure with proven 0-to-1 product launches. Currently integrating frontier LLMs models into AI-assisted workflows for production, optimizing developer velocity by 40%.

## EXPERIENCE

**Software Engineer I • Pathao Ltd.** *Dec'24 - Present*

Full-stack engineer delivering scalable platforms across mapping, campaign management, and real-time systems serving 100K+ daily users while contributing to Super AI team's LLM integration efforts.

- **Dashboard Foundation:** Engineered NextJs framework using ReactJs best practice with OpenAPI-driven type generation and Zod validation, powering rapid cross-sector development with centralized auth/RBAC, and embedded AI guardrails.
- Pathao Map (Alexander): Architected microservices-based community platform with RBAC for gathering, validating POIs serving 100K+ daily users using • GoLang • VueJs • NuxtJs • Docker.
- **Loop:** Built campaign management platform handling marketing operations across all services with bulk **XLSX uploads** and robust file handling and real-time analytics. Optimized system stability and error feedback.
- **Mission:** Developed gamification platform with event-based listeners rewarding users for action streaks. Built dashboard for mission, entry, phase, goal management andanalytics using • NextJs • ReactJs• Go • PostgreSQL • Redis • BigQuery.
- Resto Web: Engineered restaurant management SaaS with WebSocket real-time notifications, dynamic menus, analytics dashboard using • NextJs • ReactJs • Zod • TypeScript.
- Year End Wrap: Engineered interactive 2025 wrap experience integrated without app updates. Optimized assets and solved critical cross-platform (iOS/Android) compatibility issues using • NextJs • ReactJs • TypeScript.
- **GariKoi:** Developed real-time geo tracking system visualizing driver locations, allocation/deallocation for operations.

**Associate Software Engineer • ShellBeeHaken Ltd.** *Aug'23 - Nov'24*

Promoted to full-time after delivering production applications across diverse business domains.

- Mapage: Built real-time communication system for reservation/payment platform using • Socket.IO • NextJs • ReactJs • MongoDB with optimistic updates and conflict resolution.
- UUORK: Engineered cross-platform workspace app (ElectronJs + NextJs) with JWT authentication, real-time collaboration, responsive design.
- Honnemir: Developed multimedia voting platform with • GraphQL API • AWS Amplify • S3 upload pipeline • admin moderation dashboard.
- KriyaKarak: Implemented payment disbursement system with automated payouts • reconciliation dashboard for creative marketplace using • NextJs • ReactJs • MongoDB.

**Software Engineer Intern • ShellBeeHaken Ltd.** *Feb'23 - Aug'23*

Shipped features and fixed critical bugs across multiple production codebases with agile processes.

## PROJECTS

**Obstacle Detection**
Accessibility app using TensorFlow Lite for real-time object detection with audio warnings for visually impaired.
• Java • TensorFlow Lite • Computer Vision

**Flash Nvim for VSCode**
Published extension porting flash.nvim navigation to VS Code with 1500+ installs (also on **Open VSX** for VSCodium/Cursor), 5-star rating, intelligent search, multi-cursor. • TypeScript

**Sync Play**
Collaborative video platform with synchronized playback, chat, local/cloud media support using WebSocket real-time.
• NextJs • Socket.IO • Firebase

**FTP Searcher**
Cross-platform FTP aggregator with search indexing, content discovery for local network sharing. • NextJs • ElectronJs

## EDUCATION

**Shahjalal University of Science and Technology, Sylhet, Bangladesh** *Jan'19 - Jan'24*

Institute of Information and Communication Technology (IICT)
**Bachelor of Science in Software Engineering • CGPA: 3.64/4.00** • Ranked 2nd in class (all courses first attempt)

## PUBLICATIONS

IPA Transcription of Bengali Texts (arXiv 2024) — Standardized Bengali IPA phonetic transcription with novel dataset and deep learning model advancing NLP for low-resource languages. • Python • TensorFlow • NLP • Pandas

## DATASETS

**DUALIPA:** A comprehensive Bengali IPA dataset featuring 156k+ validated sentences with phonetic transcriptions generated via deep learning models and expert verification. • Literature (103k) • Newspaper (53k)

**Regional IPA:** Covering phonetic variations across 7 major regions (Rangpur, Kishoreganj, Chittagong, Narsingdi, Narail, Habiganj, Tangail) with sentence and word-level transcriptions for dialectal analysis. • Dataset